

A telescoping proof of the focal antipedal sum invariant

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6 October 2026 — Version 1.0

Abstract

Reznik, Garcia and Koiller observed that the sums of the distances from the two foci of an ellipse to their respective antipedal vertices agree along every closed elliptic billiard orbit. We prove this equality for every regular closed orbit tangent to a strictly nested confocal ellipse, including odd primitive periods, star orbits and repeated traversals. The difference of the two ordinary positive antipedal distances on each edge is a fixed coefficient times the vertical displacement of that edge. Closure therefore gives the equality without a parity restriction.

1 Statement, construction and attribution

Let

$$E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad E_\lambda : \frac{x^2}{a^2 - \lambda} + \frac{y^2}{b^2 - \lambda} = 1, \quad a > b > 0, \quad 0 < \lambda < b^2.$$

Put $c = \sqrt{a^2 - b^2}$ and $f_\sigma = (\sigma c, 0)$ for $\sigma \in \{+1, -1\}$. A regular closed billiard/Poncelet orbit here is a cyclic list $P_1, \dots, P_N$ on E , with distinct consecutive endpoints, whose edges are tangent to E_λ and whose continuation at each vertex uses the other of its two caustic tangents. Thus immediate retracing is excluded. The list may self-intersect, have any admissible winding, or repeat a shorter orbit. Set $P_{N+1} = P_1$.

For a focus f_σ , define $Q_{\sigma,i}$ as the intersection of the full lines through P_i and P_{i+1} perpendicular to $P_i - f_\sigma$ and $P_{i+1} - f_\sigma$, respectively. These are the vertices of the antipedal of the original orbit polygon about that focus. We use ordinary Euclidean distances $q_{\sigma,i} = \|Q_{\sigma,i} - f_\sigma\|$.

Theorem 1. *Every regular closed orbit just defined has finite, strictly positive antipedal distances and satisfies*

$$\sum_{i=1}^N q_{+,i} = \sum_{i=1}^N q_{-,i}. \tag{1}$$

In particular the ratio of these two sums is 1 for every member of every admitted Poncelet family.

This is the observation labelled k_{603} by Reznik–Garcia–Koiller in [1, §§3.5,3.7 and Table 7] and [2, §3.7 and Table 7, p. 349]. Their table records its experimental discovery in May 2020 and marks its proof as unknown there. That dated marker is not a claim about the present literature. The observation and antipedal construction are theirs, and the underlying negative-pedal/polarity geometry is classical [3, Arts. 121–122]. The focal sideline-height product used below is also known [4, Lemma 4.2]. Central inversion already proves the equality for centrally paired orbits; ordinary telescoping is a familiar method. The contribution proposed here is the identification of the ordinary norm difference with a coordinate increment and the resulting full-period proof, in particular for odd primitive periods.

Neither individual sum is asserted constant as an orbit varies. The claim concerns the original polygon, not the outer tangent polygon, pedal feet, side perimeters or signed radial lengths. Hyperbolic caustics, degenerate caustics and two-bounce limits are outside its hypotheses.

2 Positive antipedal distance and orientation

Lemma 1. *For noncollinear A, B, f , the antipedal intersection Q is unique and*

$$\|Q - f\| = \frac{\|A - f\|\|B - f\|}{h}, \quad h = \text{dist}(f, \text{line}(A, B)) > 0. \quad (2)$$

Proof. Translate f to zero and write $A = (r, 0)$, $B = (s \cos \alpha, s \sin \alpha)$ with $r, s > 0$ and $\sin \alpha \neq 0$. The defining lines give $Q \cdot A = r^2$, $Q \cdot B = s^2$, hence

$$\|Q\|^2 = \frac{r^2 + s^2 - 2rs \cos \alpha}{\sin^2 \alpha} = \frac{\|A - B\|^2}{\sin^2 \alpha}.$$

The triangle area identity gives $h = rs|\sin \alpha|/\|A - B\|$. Taking the positive square root proves (2). $\square$

The whole orbit can be oriented with the caustic on the left of each directed edge. Indeed, a tangent line to the strictly inner ellipse meets E in two points, and its contact lies strictly between them. At a vertex the rays toward the two contact points bound a cone containing the caustic, so the caustic lies on opposite sides of those two directed rays. Reversing the incoming ray and continuing on the other tangent therefore preserves its side. The nonretracing rule propagates one consistent side throughout the orbit; reverse the list if it is the right side. This is a local argument and does not require a convex polygon or any ordering of its vertices around E .

For such a directed edge AB , take its outward unit normal $n = (u, v)$ and tangent $t = (-v, u)$. Its equation is $n \cdot X = \rho$, with $u^2 + v^2 = 1$ and $\rho > 0$; the caustic and origin lie on the side $n \cdot X < \rho$. Put

$$K = a^2 u^2 + b^2 v^2, \quad \rho^2 = K - \lambda. \quad (3)$$

The second equality is the support-function condition for tangency to E_λ . The chord endpoints are

$$A = M - \ell t, \quad B = M + \ell t, \quad M = \left(\frac{a^2 \rho u}{K}, \frac{b^2 \rho v}{K} \right), \quad \ell = \frac{ab\sqrt{\lambda}}{K} > 0. \quad (4)$$

To check this, substitute $M + zt$ in the outer ellipse equation: the linear term vanishes, the constant is ρ^2/K , and the quadratic coefficient is $K/(a^2 b^2)$. Thus $z = \pm \ell$.

The two focal heights are

$$h_\sigma = \rho - \sigma c u, \quad h_+ h_- = \rho^2 - c^2 u^2 = b^2 - \lambda > 0. \quad (5)$$

Their sum is $2\rho > 0$, so both heights are strictly positive. They are therefore the ordinary perpendicular distances of the foci to the line, not signed substitutes. Equivalently, both foci lie strictly inside E_λ , since $c^2 < a^2 - \lambda$. No edge is focal, and Lemma 1 applies to both foci.

3 The edge identity and closure

For a point (x, y) on E , its distance to f_σ is $a - \sigma c x/a$: squaring verifies the identity, and this expression is at least $a - c > 0$. From (4),

$$x_A + x_B = \frac{2a^2 \rho u}{K}, \quad x_A x_B = \frac{a^2(\rho^2 - b^2 v^2)}{K}.$$

Multiplying the two positive endpoint distances gives

$$R_\sigma := \|A - f_\sigma\| \|B - f_\sigma\| = \frac{a^2 h_\sigma^2 + b^2 \lambda}{K}. \quad (6)$$

For completeness, the product before simplification is $a^2 - \sigma c(x_A + x_B) + (c^2/a^2)x_A x_B$; inserting the preceding two expressions and $\rho^2 = K - \lambda$ yields (6). Lemma 1 now gives positive distances

$$q_\sigma = \frac{1}{K} \left(a^2 h_\sigma + \frac{b^2 \lambda}{h_\sigma} \right). \quad (7)$$

Since $h_+ - h_- = -2cu$ and $h_+^{-1} - h_-^{-1} = 2cu/(b^2 - \lambda)$, subtraction gives

$$q_+ - q_- = \frac{2cu}{K} \left(-a^2 + \frac{b^2\lambda}{b^2 - \lambda} \right). \quad (8)$$

But (4) also gives $y_B - y_A = 2ab\sqrt{\lambda}u/K$. Hence

$$\boxed{q_+ - q_- = \Gamma(y_B - y_A)}, \quad \Gamma = \frac{c}{ab\sqrt{\lambda}} \left(-a^2 + \frac{b^2\lambda}{b^2 - \lambda} \right). \quad (9)$$

All denominators are positive. No division by u or by the vertical displacement occurs, so horizontal chords are included. The coefficient vanishes at $\lambda = a^2b^2/(a^2 + b^2)$ and changes sign there; neither fact affects positivity of q_σ . Formula (9) uses the chosen caustic-on-left orientation. Reversing endpoints leaves $q_+ - q_-$ unchanged but negates $y_B - y_A$; in the reverse orientation its displacement coefficient is $-\Gamma$.

Proof of Theorem 1. Use the consistent left orientation above. The same λ and Γ apply to every edge. Summing (9) gives

$$\sum_i (q_{+,i} - q_{-,i}) = \Gamma \sum_i (y_{i+1} - y_i) = 0$$

by closure. Equations (5) and (2) show that every summand is positive and finite, so the denominator of the ratio is nonzero. No parity condition is used. Repeated traversal simply repeats the summands. Reversing an edge leaves its two antipedal intersections unchanged, hence reversing the orbit preserves the equality. $\square$

4 Verification and status

The accompanying standard-library Python verifier independently solves the two antipedal line equations on rational chords. It checks the norm, positive focal heights, confocal tangency and the edge identity exactly; it also covers horizontal chords, zero and opposite signs of Γ , reversal and a closed repeated four-cycle. Deliberately incorrect coefficient, height, norm sign and manuscript-hash controls must fail. All guards remain active under Python optimization. Its recorded counts are in the support files. These finite computations supplement the written proof for all real parameters and every admitted period.

A bounded primary-source and mechanism audit completed on 6 October 2026 located no earlier full covering proof. In particular, focal inverse perimeters [5] concern different summands, and the complete journal final [6], compared with its precursor, provides spatial averages for other observables rather than this finite antipedal norm difference. This is not an absolute-first or continued-open-status guarantee. The support audit summary retains the explicit final-book versus author-source boundary, unread historical and experimental materials, and finite search/version limits.

Disclosure. AI tools were used extensively in solving, deriving, drafting, literature checking, code generation, verification and adversarial checking. The independent checks described in the support package include AI agents and are not conventional human peer review. This manuscript is unrefereed and has not received conventional human peer review. The manuscript and original support files are licensed under [CC BY 4.0](#); cited third-party works retain their own terms and are not redistributed.

References

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